

ANSWER-1

**AI in Data Visualization** means using artificial intelligence to automatically understand patterns, trends, and relationships in data and then present them visually. Instead of manually exploring charts, AI highlights important insights for you.

**it differs from traditional analysis:**

- **Traditional analysis** is manual — you choose charts, explore patterns, and interpret results yourself.
- **AI-based visualization** is automatic — AI scans the entire dataset, finds hidden patterns, predicts trends, and explains insights without you doing manual exploration.

ANSWER-2

**AI helps BI tools automatically generate insights by studying the entire dataset and identifying patterns without manual effort.** Instead of the user checking each chart one by one, AI scans the data and highlights what matters most.

ANSWER-3

Some important AI features are:

- Key Influencers
- Smart Narratives
- AI Visuals
- Auto ML (Automated Machine Learning)
- Q&A (Ask questions in natural language)
- Decomposition Tree
- Anomaly Detection

### **1. Key Influencers**

**What it does:** Key Influencers uses AI to find which factors have the strongest impact on a selected metric.

**How it works:** Power BI scans all columns in the dataset and identifies what increases or decreases the value you are analyzing.

**Example:** If you choose “**Sales Amount**” as the metric, Key Influencers may show:

- “High discount leads to higher sales.”
- “Returning customers buy more than new customers.”

### Smart Narratives

**What it does:** Smart Narratives automatically generates text summaries for your visuals and dashboards.

**How it works:** It reads your charts and writes meaningful insights in simple language.

**Example:** If your dashboard shows quarterly sales, Smart Narratives may generate:

- “Sales increased by 12% this quarter.”
- “North region contributed the highest growth.”

ANSWER-4

### Concept of Key Influencers

Key Influencers is an AI-powered visual in Power BI that helps you understand **why** a particular metric changes. It automatically analyzes your dataset and shows which factors have the strongest impact on the value you are studying.

In simple words: **Key Influencers tells you what drives a metric up or down.**

### How AI identifies factors affecting a metric

AI uses statistical models and machine learning to scan all the columns in your dataset. It then:

- **Checks relationships** between the selected metric and other fields
- **Finds patterns** that increase or decrease the metric
- **Measures the strength** of each factor
- **Compare different groups** to see what influences the outcome
- **Highlights top contributors** that matter the most

ANSWER-5

### Role of Smart Narratives

- **Creates automatic summaries** It converts charts and numbers into easy-to-understand sentences.

- **Highlights key trends and changes** Smart Narratives points out increases, decreases, and important patterns in the data.
- **Explains insights in simple language** Even non-technical users can understand what the dashboard is showing.
- **Saves analysis time** You don't have to manually write observations — AI does it instantly.
- **Makes dashboards more interactive** Users can add dynamic text boxes that update automatically when data changes.

#### ANSWER-6

**Tableau Pulse** is an AI-powered feature that gives users personalized, real-time insights without needing to open dashboards. It focuses on delivering the *right information at the right time* by continuously monitoring important metrics.

**Tableau Pulse provides proactive insights:**

- **Automatically tracks key metrics** Pulse keeps an eye on important business numbers like sales, customer retention, or revenue.
- **Sends alerts when something unusual happens** If a metric suddenly drops or increases, Pulse notifies you immediately.
- **Provides natural-language explanation**
  - It explains insights in simple sentences, so even non-technical users can understand what changed and why.
  - Gives **personalized insights for each user** Every user receives updates based on the metrics they care about — not generic dashboard information.
  - **Highlights trends and patterns** Pulse identifies what is driving changes, such as region performance, customer behavior, or product demand.

#### ANSWER-7

AI-driven decision-making means using artificial intelligence to help businesses make smarter and faster decisions. Instead of relying only on human judgment, AI analyzes large amounts of data, finds patterns, and suggests the best possible actions.

In simple words: **AI helps businesses decide “what to do next” by studying data automatically.**

AI supports decision-making in several ways:

- **Predicts future trends** AI models can forecast sales, demand, customer behavior, and risks.
- **Finds hidden patterns** It identifies relationships in data that humans may not notice.
- **Provides real-time insights** AI updates decisions instantly based on new data.
- **Recommends best actions** Example: AI may suggest increasing stock for a product because demand is rising.
- **Reduces human errors** Decisions become more accurate because they are based on data, not guesswork.
- **Helps in quick decision-making** Businesses can act faster because AI processes information instantly.

ANSWER-8

### **AI depends heavily on data quality**

If the data is incomplete, incorrect, or biased, AI will generate wrong insights. AI cannot fix bad data — it only analyzes what it receives.

### **2. AI may misinterpret business context**

AI does not fully understand real-world situations or business logic. It may highlight patterns that are statistically correct but practically irrelevant.

### **3. Insights can sometimes be misleading**

AI might show correlations that look important but have no real meaning.

Example: “Sales increased when temperature increased” — but this may not be the true cause.

### **4. Limited ability to explain complex reasoning**

AI can show patterns, but it cannot always explain *why* something is happening in deep detail. Human expertise is still needed for final interpretation.

### **5. Over-reliance can reduce human analysis skills**

If users depend too much on AI insights, they may stop exploring data manually, missing deeper insights.

### **6. Requires technical setup and good models**

AI features need proper configuration, training data, and model selection. Without this, results may be inaccurate or incomplete.

## 7. Cannot handle emotional or strategic decisions

AI works on numbers, not emotions, customer behavior, or long-term strategy. Businesses still need human decision-makers.

ANSWER-10

## AI Approach

- **Power BI:** Focuses on *user-friendly AI* that automatically explains insights.
- **Tableau:** Focuses on *advanced analytics* and deeper statistical explanations.

## 2. AI Features

- **Power BI:**
  - Key Influencers
  - Smart Narratives
  - Auto ML
  - Q&A (Ask questions in natural language)
- **Tableau:**
  - Tableau Pulse
  - Explain Data
  - Einstein Discovery (Salesforce integration)
  - Predictive modeling

## 3. Insight Style

### . Power BI:

AI generates **text-based summaries** and simple explanations.

. **Tableau:** AI provides **visual + statistical insights** with deeper analysis.

## 4. Ease of Use

. **Power BI:** Very beginner-friendly; AI insights appear automatically.

. **Tableau:** More advanced; requires analytical understanding to use AI features effectively.

## 5. Automation Level

- **Power BI:** High automation — AI writes narratives and identifies influencers instantly.
- **Tableau:** More user control — AI supports analysis but doesn't fully automate everything.